WEATHORA

Weather Dashboard with API

Project Presentation
Web Development Internship

Submitted By: Sanika Sanjay Dhavale
B.Sc. Computer Science-Final year
MES Abasaheb Garware College, Pune

Project Overview

Weathora-Weather Dashboard with API

Weathora is a web-based application that allows users to check the current weather of any city.

Main Objectives:

- To fetch live weather data using an API.
- To display weather details in a user-friendly interface.
- To store search history in MongoDB.
- To allow users to view their previous city searches.

Problem Statement & Objectives

Problem Statement:

Checking weather information for different cities can be time-consuming when users need to visit different websites.

Objectives:

- To provide real-time weather information for any city.
- To display temperature, humidity, and wind speed.
- To store city search history using MongoDB.
- To create a simple and user-friendly web application.

Technology Stack

- Frontend: HTML, CSS, JavaScript
- Backend: Python, Flask
- Weather Data: OpenWeather API
- Database: MongoDB
- Development Tool: Visual Studio Code

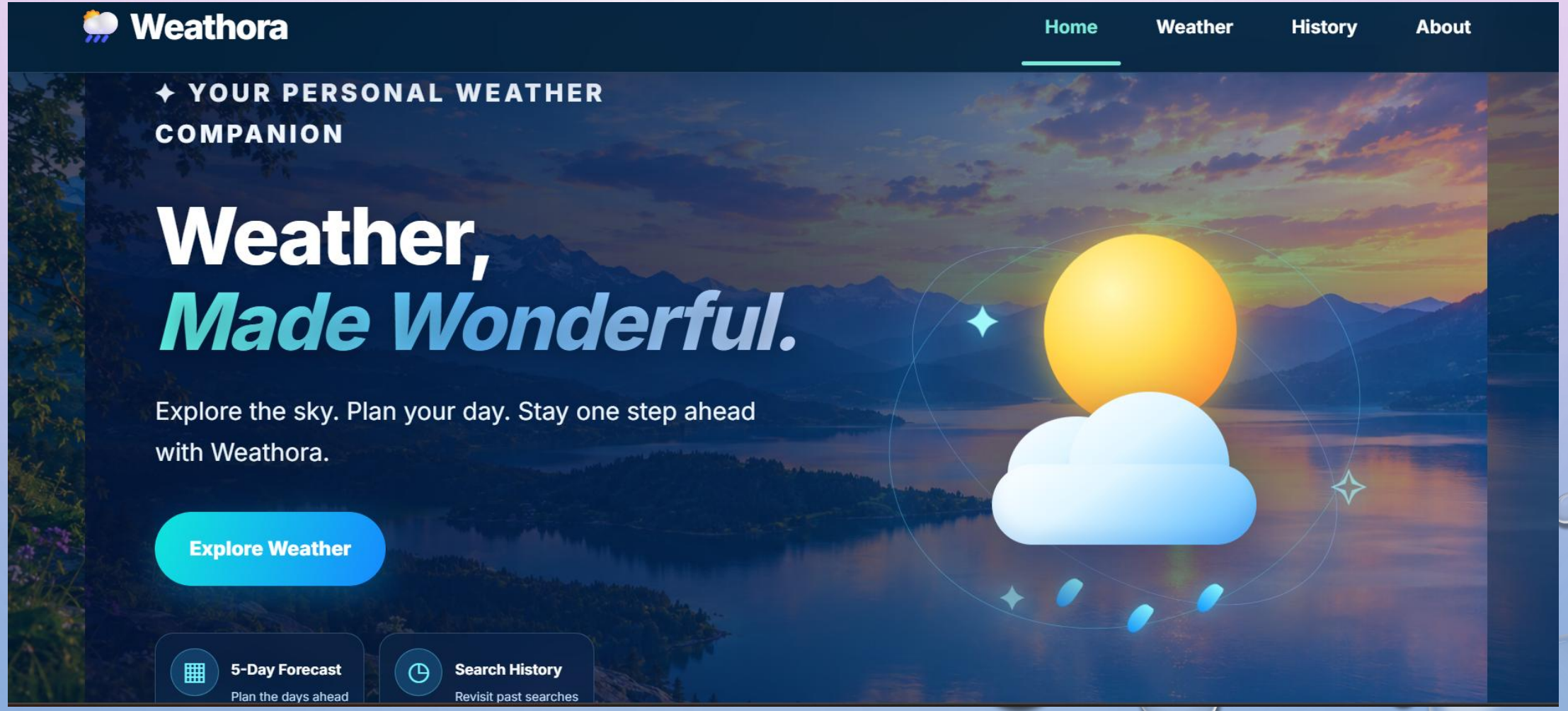
System Workflow

- User enters a city name.
- Flask processes the request.
- OpenWeather API fetches current weather data.
- MongoDB stores the search history.
- Weather results are displayed on the website.

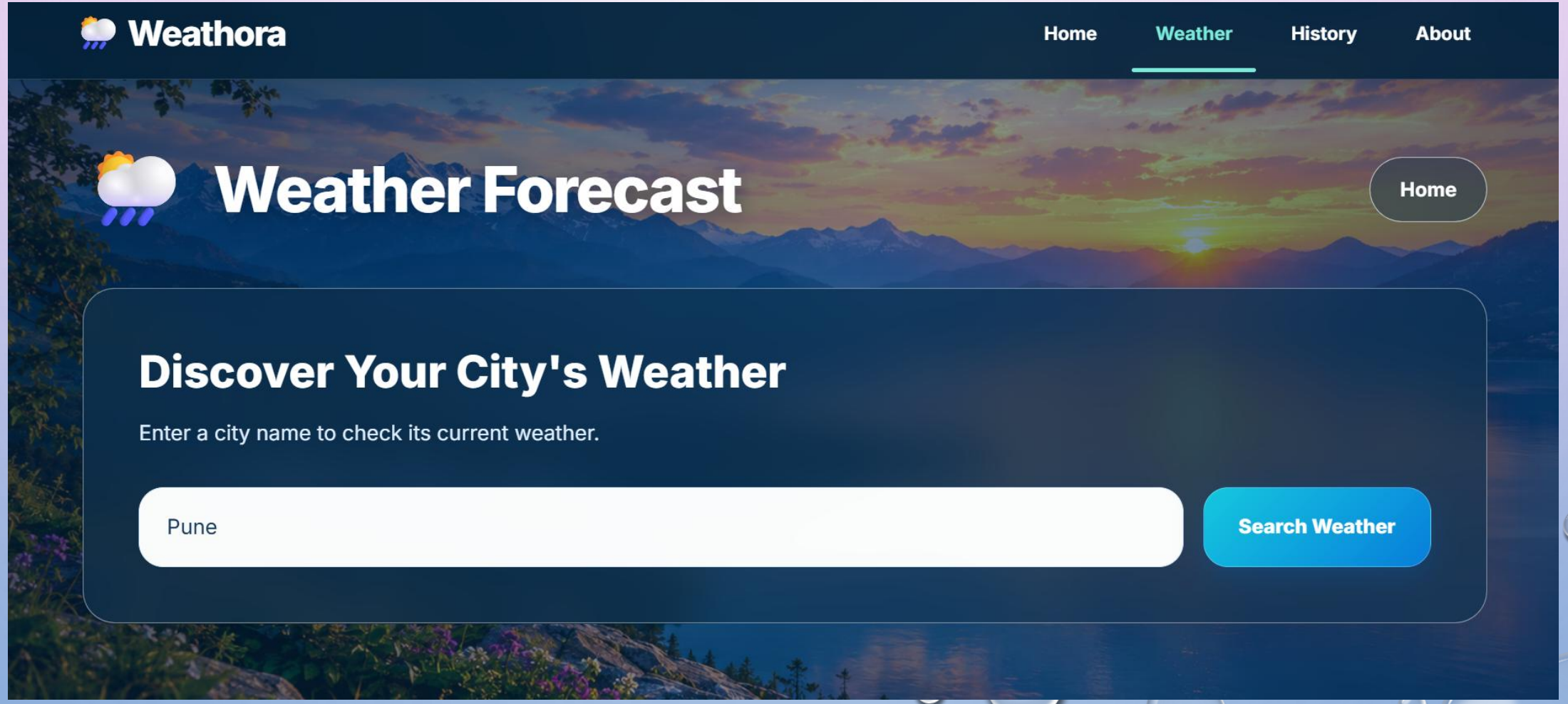
Main Features of Weathora

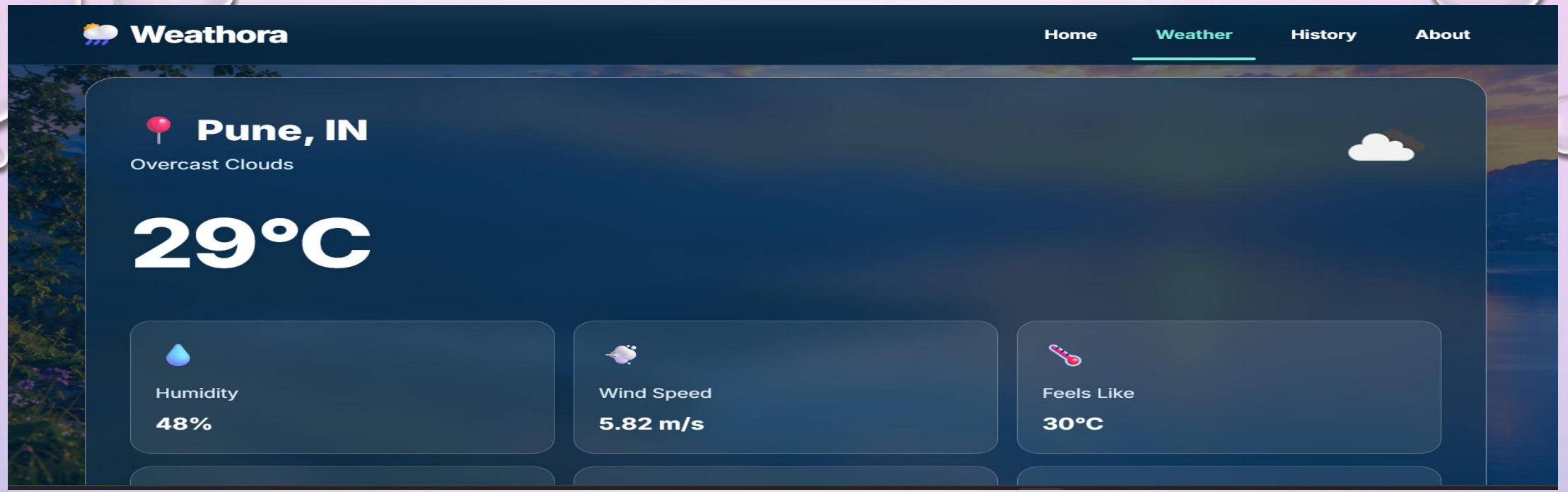
- Weather information using OpenWeather API.
- Temperature, humidity and wind speed details.
- Search history stored in MongoDB.
- User-friendly and responsive interface.
- Separate Home, Weather, History and About pages.

Home Page


Weathora
Home
Weather
History
About
✦ YOUR PERSONAL WEATHER COMPANION
Weather,
Made Wonderful.
Explore the sky. Plan your day. Stay one step ahead with Weathora.
Explore Weather
5-Day Forecast
Plan the days ahead
Search History
Revisit past searches

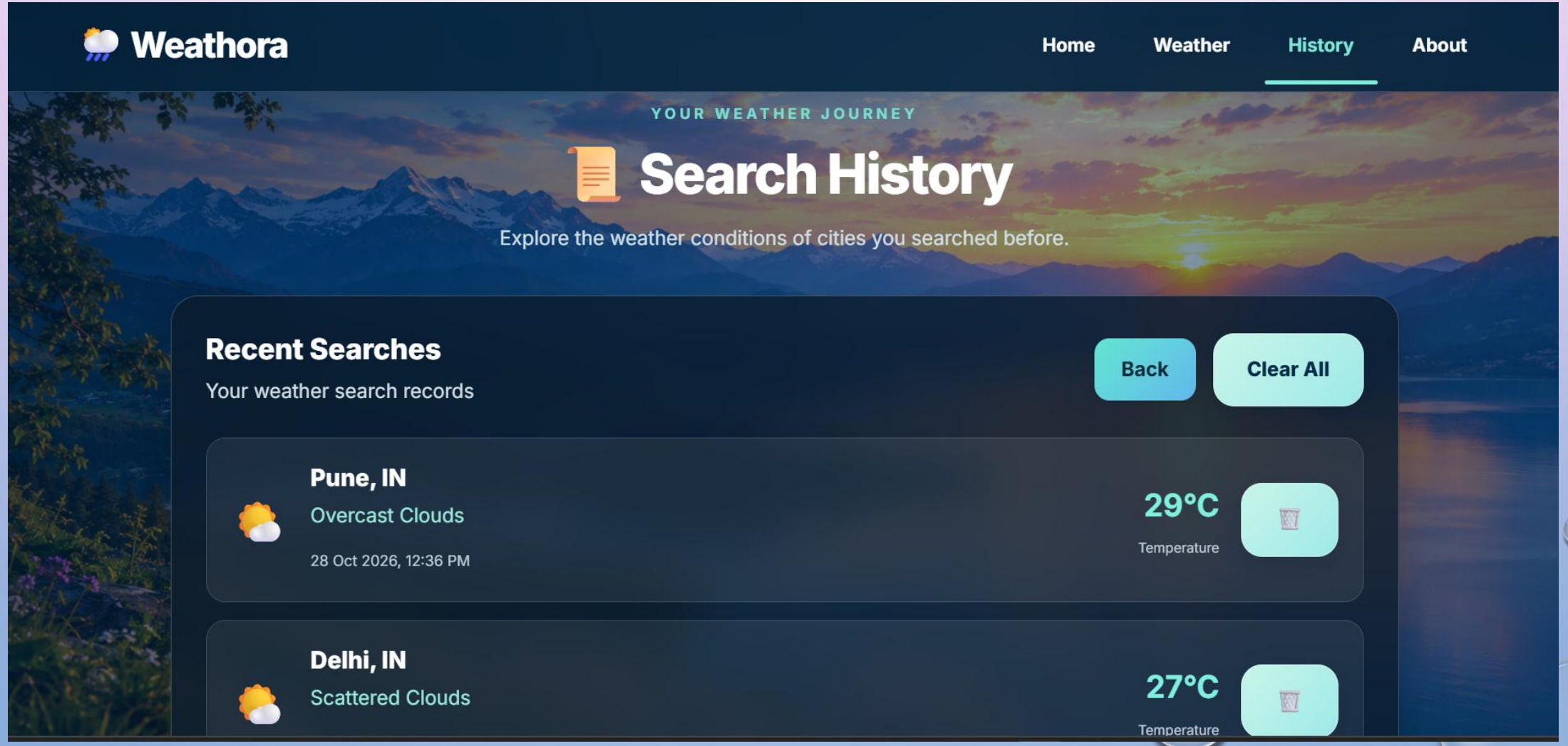
Weather Forecast Page




Weathora
Home
Weather
History
About
Pune, IN
Overcast Clouds
29°C
Humidity
48%
Wind Speed
5.82 m/s
Feels Like
30°C


Weathora
Home
Weather
History
About
Pressure
1008 hPa
Visibility
10.0 km
Sunrise
06:32 AM
Next 5 Days Forecast
Wed
28 Oct
29°C
Overcast Clouds
Thu
29 Oct
31°C
Scattered Clouds
Fri
30 Oct
27°C
Broken Clouds
Sat
31 Oct
29°C
Clear Sky
Sun
01 Nov
28°C
Broken Clouds

Search History Page



About Page


Weathora
Home
Weather
History
About
GET TO KNOW US
About Weathora
Your world, your weather, your way. Discover current conditions and upcoming forecasts with a simple and beautiful weather experience.
Explore Weather

OpenWeather API Integration

How It Works:

1. The user enters a city name.
2. Flask sends a request to the OpenWeather API.
3. The API returns weather data for the selected city.
4. Weathora displays the weather information on the website.

Weather Data Retrieved:

- City name
- Temperature
- Humidity
- Wind speed
- Feels-like temperature

MongoDB Integration & Testing

MongoDB Integration:

- Stores users city search history.
- Saves search records in the database.
- Retrieves previous searches for the history page.
- Uses Flask to connect the application to MongoDB.

Testing Performed:

- Checked weather search for different cities.
- Verified weather details displayed on the page.
- Checked whether search history was saved.
- Tested the history page.

Conclusion & Future Enhancements

Conclusion:

Weathora is a web-based weather dashboard developed using Python, Flask, the OpenWeather API and MongoDB.

- Provides current weather information for different cities.
- Displays temperature, humidity, and wind speed.
- Stores search history in MongoDB.
- Offers a simple and user-friendly interface.

Future Enhancements:

- Add a multi-day weather forecast.
- Include weather maps and alerts.
- Improve mobile responsiveness.
- Add more weather-related features.